

HOW TO READ & TRIAGE SECURITY ALERTS

FOR SOC ANALYSTS



Learn how to read, understand and triage security alerts like a SOC Analyst.

DETECT FASTER. INVESTIGATE BETTER. RESPOND SMARTER.

WHAT YOU'LL LEARN

- ✓ Understand alerts like a pro
- ✓ Identify real threats vs false alarms
- ✓ Use a structured triage framework
- ✓ Make smart decisions with confidence
- ✓ Improve SOC efficiency & response time

“ The best analysts aren't just good at tools, they're good at THINKING. ”



2. WHAT IS A

SECURITY ALERT?



A security alert is a warning message generated by a **security tool** to inform us about **potentially suspicious** or **malicious activity** in our environment.

WHY ARE ALERTS GENERATED?

- ✓ To detect threats and **unusual behavior**
- ✓ To protect systems, data and users
- ✓ To help analysts respond quickly
- ✓ To reduce the impact of attacks



WHERE DO ALERTS COME FROM?

Alerts can come from many security tools such as:



EDR (Endpoint Detection & Response)



SIEM (Security Information & Event Management)



Firewall



IDS/IPS



Cloud, Email, Identity, Applications and more...

REAL EXAMPLES

- **Multiple Failed Login Attempts** → Someone may be trying to brute force.
- **Malware Detected** → A malicious file was found and blocked.
- **Suspicious PowerShell Execution** → A command was executed that looks dangerous.
- **Impossible Travel** → User logged in from two distant locations.
- **Data Exfiltration Detected** → Possible data is being sent outside the network.



KEY TAKEAWAY



Not every alert is an attack, but **every attack starts with an alert.**

Our job as SOC Analysts is to **read, understand** and decide the **right action!**



Think of an alert as a **clue**. Your job is to follow the clue and find out the **truth behind it!**



3. WHAT IS

ALERT TRIAGE?



Alert triage is the process of **reviewing**, **prioritizing** and **deciding** what action to take on a security alert.

THINK OF IT LIKE THIS...

Just like doctors in an **emergency room...**

- Many patients (alerts) arrive at the same time.
- Some are **critical**, some are not.
- Doctors check, prioritize and treat the right patient first.

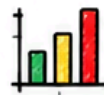


WHAT DOES TRIAGE MEAN?



REVIEW

Look at the alert carefully.



PRIORITIZE

Decide how important it is.



DECIDE

Figure out what it means.



TAKE ACTION

Escalate, investigate, contain or close.

WHY DO SOC ANALYSTS TRIAGE ALERTS?



Focus on real threats that matter.



Save time and respond faster.



Reduce risk to the organization.



Avoid alert overload and stress.



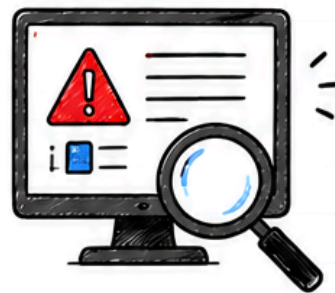
Improve SOC efficiency and accuracy.



Good triage = **Right alert**, **Right time**,
Right action!



4. HOW SOC ANALYSTS READ ALERTS?



Reading an alert is not just looking at it.
It's **understanding** the **story** behind it.

STEPS TO READ AN ALERT EFFECTIVELY

1



READ THE ALERT TITLE

- The title tells you what happened.
- Get the **first impression**.

2



CHECK THE SEVERITY

- See how serious the alert is.
- Helps you decide **priority** and **urgency**.

3



CHECK THE AFFECTED ASSETS

- Which host, user or system is involved?
- Know your **environment**.

4



REVIEW THE EVIDENCE

- Look at IPs, domains, files, processes, commands, logs.
- These are the **clues**.

5



CHECK THE TIMELINE

- When did it happen?
- Look for **patterns**, sequence and related events.

ALWAYS REMEMBER:



- ★ Read everything **carefully**.
- ★ Don't **assume** anything.
- ★ Use **context**, not just alert text.
- ★ Think like an **attacker**.
- ★ **Validate** before you escalate.

EXAMPLE ALERT (SAMPLE)



Alert: Suspicious PowerShell Execution



Host: WIN-10-23



User: john.doe



Time: 2025-05-12 10:43:21



Source IP: 185.199.108.153



Severity: **High**



The better you **read**,
the better you **respond**!



5. WHY ALERT TRIAGE MATTERS?



Good triage
= Strong
Security!



SOC teams receive **thousands of alerts** every day. You cannot investigate everything. Triage helps you focus on **what really matters**.

KEY REASONS WHY ALERT TRIAGE IS CRITICAL



1. TOO MANY ALERTS EVERY DAY

SOC tools generate a huge number of alerts. Triage helps you **separate important from noise**.



2. REAL ATTACKS CAN HIDE

Attackers hide in the noise. Poor triage can cause you to **miss real threats**.



3. FASTER TRIAGE = FASTER RESPONSE

Quick triage helps you respond early and **reduce the impact** of an attack.



4. PROTECTS THE ORGANIZATION

Good triage leads to better decisions, stronger security and **protects sensitive data**.



5. IMPROVES YOUR SKILLS

The more you triage, the **better you understand** attacks, tools and attacker behavior.



Good triage saves time, reduces risk and helps your team **stay ahead of attackers!**



6. COMMON TYPES OF SECURITY ALERTS



Security tools monitor your environment 24/7 and generate alerts when something **suspicious** happens.

HERE ARE SOME COMMON ALERT TYPES:



1 FAILED LOGIN ATTEMPTS

Multiple failed attempts to login to an account.
May indicate brute force or credential stuffing.



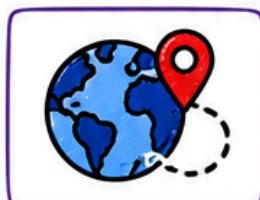
2 MALWARE DETECTION

Malicious file or behavior detected on a system.
Could be a virus, trojan, ransomware, etc.



3 SUSPICIOUS POWERSHELL ACTIVITY

Abnormal PowerShell commands or scripts executed.
Often used by attackers.



4 IMPOSSIBLE TRAVEL

User logged in from two distant locations in a short time.
May indicate account compromise.



5 RANSOMWARE ACTIVITY

File encryption, ransom notes, or unusual file changes.
Critical alerts!



6 PRIVILEGE ESCALATION

User gained higher access than normal.
Could lead to full system compromise.



Not all alerts are attacks,
but all attacks start with an alert.



7. WHAT MAKES AN ALERT SUSPICIOUS?



Not every alert is dangerous.

Certain **signs** or **patterns** make an alert **suspicious** and worth investigating.

KEY THINGS THAT MAKE AN ALERT SUSPICIOUS:



1 UNUSUAL BEHAVIOR

- Activity that is not normal for the user or system.
- Example: User logs in at **3 AM** from a **new country**.



2 UNKNOWN OR SUSPICIOUS IPs

- Alerts involving IP addresses that are unknown or blacklisted.
- Example: Connection from a **high-risk IP**.



3 RARE OR UNCOMMON PROCESSES

- Processes or commands that are rarely used.
- Example: Suspicious **PowerShell** or unknown executable.



4 AFTER-HOURS ACTIVITY

- Important systems or accounts active outside business hours.
- Example: Admin login at **midnight**.



5 MULTIPLE FAILED ATTEMPTS

- Many failed logins, file access attempts, or permission errors.
- Example: **10+ failed logins** in 5 minutes.



6 UNUSUAL FILE CHANGES

- Critical files modified, deleted, or encrypted.
- Example: **Security settings** changed unexpectedly.



KEY TAKEAWAY: When activity is **unusual**, **uncommon**, or **unexplained**, it's worth a closer look!



8. WHAT IS A TRUE POSITIVE?



A TRUE POSITIVE means the alert is **CORRECT** and indicates real **malicious activity** or a real threat.

In short: The alert is **real** and **needs action**.

KEY POINTS ABOUT TRUE POSITIVES:



REAL THREAT

Confirms malicious activity or attacker behavior.



NEEDS RESPONSE

Requires immediate or timely investigation and action.



EVIDENCE EXISTS

There is clear evidence or indicators supporting the alert.



HIGH RISK

May impact systems, data, users or the organization.

EXAMPLES OF TRUE POSITIVES:

1



REAL MALWARE DETECTION

Malware file executed on a system and confirmed malicious by analysis tools.

2



REAL PHISHING ATTACK

User clicked a malicious link and entered credentials on a fake website.

3



REAL BRUTE FORCE ATTACK

Multiple failed login attempts from an attacker IP trying to gain access.



KEY TAKEAWAY:

A true positive is a **real threat**.
Investigate, escalate and take appropriate action.



9. WHAT IS A FALSE POSITIVE?



A FALSE POSITIVE means the alert looks suspicious, but it is actually NOT malicious. It is a benign or normal activity.



In short: The alert is NOT a real threat.

KEY POINTS ABOUT FALSE POSITIVES:



NOT MALICIOUS

The activity is not harmful or not related to an attack.



NORMAL ACTIVITY

It can be a legitimate user, process, or system behavior.



TRIGGERS ALERT

The tool or rule triggers an alert even though there is no real threat.



NO ACTION NEEDED

After review, no further action is required for the alert.

EXAMPLES OF FALSE POSITIVES:

1



LEGITIMATE ADMIN ACTIVITY

An admin uses tools or runs commands that look suspicious but are normal for their job.

2



KNOWN SOFTWARE BEHAVIOR

A software update or script triggers an alert but is a normal and expected process.

3



SECURITY SCANNER ACTIVITY

Vulnerability scans, health checks, or monitoring tools trigger alerts by design.

4



LEGITIMATE USER ACTION

A user logs in from a new location or device but has a valid business reason.



KEY TAKEAWAY:

False positives are NOT bad. They help us fine-tune our tools and rules to focus on real threats.



10. TRUE POSITIVE VS FALSE POSITIVE



Understanding the difference helps you take the RIGHT action every time!

TRUE POSITIVE (REAL THREAT)



**INDICATES A
REAL THREAT**
Malicious activity
or attacker behavior.

VS



**INDICATES
SAFE ACTIVITY**
Legitimate or
normal behavior.



**NEEDS
RESPONSE**
Investigate, contain
and take action
immediately.

VS



**DOES NOT NEED
ACTION**
Review and close
the alert.



MALICIOUS
Harmful, suspicious
or against security
policies.

VS



BENIGN
Harmless, expected
or policy-compliant
activity.



ESCALATE
May impact systems,
data, users or the
organization.

VS



CLOSE ALERT
No impact on systems
or users. No further
action needed.

REAL-WORLD EXAMPLES

TRUE POSITIVE EXAMPLES

- Malware detected and confirmed.
- Ransomware process running.
- Connection to known malicious IP.
- Unauthorized access to sensitive data.



FALSE POSITIVE EXAMPLES

- Admin runs a security tool.
- Legitimate software update.
- Vulnerability scanner activity.
- User login from a new location.



ANALYST DECISION TIPS



- ☒ Never assume – Investigate all alerts before deciding.
- ☒ Look at the full context – User, system, time, and activity.
- ☒ Use evidence – Logs, indicators and threat intelligence.
- ☒ When in doubt – Escalate, don't ignore.



DON'T DECIDE TOO QUICKLY. GOOD TRIAGE = RIGHT DECISIONS!

11. 5 KEY PIECES MOST ALERTS CONTAIN



Every security alert you see is built from a few key pieces of information. Learning to read these pieces is the **foundation of good triage!**

1 WHAT TRIGGERED THE ALERT?

What event, rule or behavior caused the alert to fire?



2 WHO / WHAT WAS AFFECTED?

Which user, host, device or system was involved?



3 WHEN DID IT HAPPEN?

What time and date did the event occur?



4 EVIDENCE / INDICATORS

What data or artifacts show that this activity was detected?



★
**MASTER THESE 5
PIECES AND YOU WILL
READ ALERTS LIKE
A PRO!**

5 SEVERITY / PRIORITY

How serious is the alert?
How quickly should it be handled?



TIP: Always look for all 5 pieces before making a decision. The more complete the picture, the better the decision!



12. WHAT TRIGGERED THE ALERT?



This piece tells you **WHAT EVENT, RULE or BEHAVIOR** caused the alert to fire.
It's the **starting point** of every investigation!

KEY THINGS TO LOOK FOR:

1 ALERT TITLE



A short summary that describes what happened. Read it carefully!

2 DETECTION RULE



The rule or signature that detected the suspicious event.

3 INITIAL IMPRESSION



A quick idea of what might be happening. Validate with facts!

4 THREAT CATEGORY



The type of threat or behavior being detected. Helps understand the risk.

EXAMPLES OF WHAT TRIGGERED THE ALERT

EXAMPLE ALERT TITLE	WHAT TRIGGERED IT?	DETECTION TYPE	THREAT CATEGORY
 Malware Detected on Endpoint	Antivirus engine detected a known malicious file.	Signature / Malware Rule	MALWARE
 Suspicious PowerShell Execution	PowerShell executed an encoded or obfuscated command.	Behavior / Heuristic Rule	POTENTIAL EXPLOIT
 Multiple Failed Login Attempts	Multiple failed logins detected from the same IP address.	Correlation / Brute Force Rule	BRUTE FORCE ATTACK

ANALYST TIPS

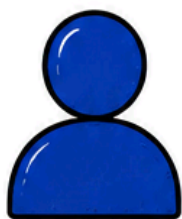
- ✓ Always read the alert **title** and **rule details** first.
- ✓ Understand **what the system saw** or detected.
- ✓ Never assume - **Validate with evidence**.
- ✓ This is the first step. **Don't jump to conclusions!**



KNOWING WHAT TRIGGERED THE ALERT HELPS YOU INVESTIGATE IN THE RIGHT DIRECTION!



13. WHO / WHAT WAS AFFECTED?



This piece tells you **WHICH USER, HOST, DEVICE** or **SYSTEM** was involved in the alert. It helps you understand the **scope** and **potential impact**.

KEY DETAILS TO CHECK:

1 HOSTNAME / DEVICE



Which device or host triggered the alert?
(e.g., PC-WS01)

2 USERNAME / ACCOUNT



Which user or account was involved?
(e.g., jsmith)

3 DEVICE TYPE



What type of device is it?
(e.g., Windows PC, Server, Mobile)

4 ASSET IMPORTANCE



How important is this asset to the organization?
(Critical, High, Medium, Low)

5 CRITICAL SYSTEMS



Is this a critical server, database or business system?

EXAMPLES

ALERT EXAMPLE	USER / ACCOUNT	HOST / DEVICE	DEVICE TYPE	IMPORTANCE
 Malware Detected on Endpoint	jdoe	PC-JDOE-15	 Windows PC	High
 Suspicious Login from New Location	asmith	SRV-APP01	 Windows Server	Critical
 Multiple Failed Login Attempts	admin	FW-EDGE-01	 Network Device	High

ANALYST TIPS

- ✓ Always identify **WHO** or **WHAT** is involved right away.
- ✓ **High value assets** need more attention.
- ✓ A **critical system** alert can have a bigger impact.
- ✓ If the affected asset is **unknown**, investigate deeper!



KNOWING WHAT WAS AFFECTED HELPS YOU FOCUS YOUR INVESTIGATION WHERE IT **MATTERS MOST!**



14.

WHEN DID IT HAPPEN?



This piece tells you **WHEN** the event occurred. **Time is critical!**
It helps you understand the **context**, identify **patterns**, and assess **urgency**.

KEY TIMING FACTORS TO CHECK:

1 TIMESTAMP



Exact date and time when the alert was generated.

2 TIME ZONE



Always check the time zone to avoid confusion.

3 BUSINESS HOURS VS AFTER-HOURS



Did it happen during normal working hours or outside them?

4 RECENCY



How recent is the alert? Is it still active or old?

TIMING CONTEXT EXAMPLES

SCENARIO	WHEN IT HAPPENED	POSSIBLE MEANING	ANALYST VIEW
 Login from new location	Mon, 10:15 AM (Business Hours)	Could be a real user working remotely.	Lower suspicion 😊
 Admin login at odd time	Sun, 2:47 AM (After-Hours)	Unusual for admin. Needs investigation.	Higher suspicion 😞
 Multiple failed logins	Over 2 hours (Repeated)	May indicate a brute force attack.	High priority !
 Alert from 2 days ago	48+ hours old	May be old noise, but still review.	Medium priority 😐

ANALYST TIPS



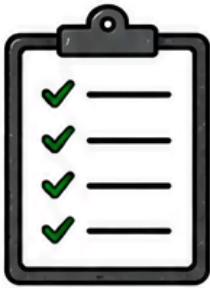
- ✓ Always check the exact **timestamp** and time zone.
- ✓ Compare the time with user's **normal behavior**.
- ✓ After-hours or weekend activity needs **more attention**.
- ✓ Don't ignore old alerts—**patterns** over time matter!



TIMING GIVES CONTEXT. CONTEXT LEADS TO **BETTER DECISIONS!**



15. EVIDENCE / INDICATORS (IoCs)



This piece contains the **EVIDENCE** or **INDICATORS (IoCs)** that show suspicious activity. These are the **clues** that help you **investigate** and make the **right decision**.

COMMON TYPES OF INDICATORS TO LOOK FOR:

1 IP ADDRESSES  Suspicious source or destination IP addresses. Example: 198.51.100.23	2 DOMAINS  Malicious or suspicious domain names contacted. Example: badguy.com	3 URLS  Suspicious web links or paths visited. Example: <code>hxxp://evil.com/payload</code>	4 FILE HASHES  Unique file fingerprints used to identify malicious files. Example: <code>ab45cd98...e67f</code>	5 PROCESS NAMES  Suspicious processes running on the system. Example: <code>powershell.exe</code>	6 COMMAND LINES  Malicious or suspicious commands executed. Example: <code>powershell -enc <base64></code>
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TOOLS THAT HELP ANALYZE INDICATORS

VIRUSTOTAL  Check files, URLs, domains, and IPs against many security engines. virustotal.com	ABUSEIPDB  Check IP address reputation and see abuse reports. abuseipdb.com	ANY.RUN  Run files and URLs in a safe sandbox to see real behavior. any.run	URLSCAN.IO  Scan and analyze URLs and websites in detail. urlscan.io
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ANALYST TIPS

- ✓ Review all **indicators** carefully—small details matter!
- ✓ Don't trust an indicator at face value—**verify it**.
- ✓ Use **multiple tools** to get a complete picture.
- ✓ Correlate indicators with other alert details and **context**.
- ✓ Good evidence leads to **accurate conclusions**!



EVIDENCE IS THE FOUNDATION OF INVESTIGATION.
ALWAYS COLLECT, VERIFY AND ANALYZE CAREFULLY!



16.

SEVERITY / PRIORITY



This piece shows **HOW SERIOUS** the alert is and **HOW URGENT** your response should be. It helps you decide what **action** to take and how **fast** to act.

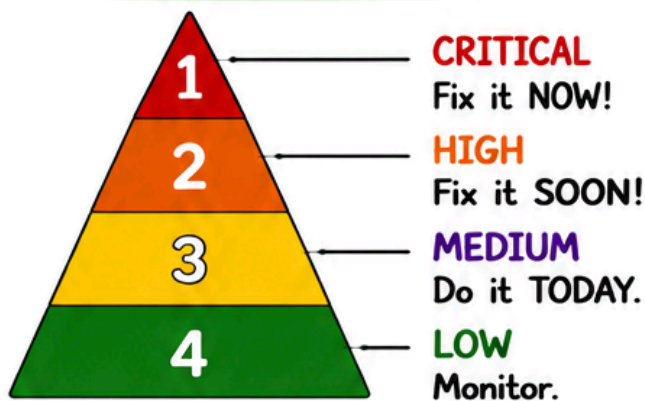
SEVERITY LEVELS EXPLAINED

SEVERITY	COLOR	WHAT IT MEANS	EXAMPLE	RESPONSE EXPECTATION
CRITICAL 	RED	Very high risk. Active attack on critical systems.	Ransomware detected on a domain controller or data exfiltration.	IMMEDIATE! Respond now.
HIGH 	ORANGE	High risk. Strong indicators of malicious activity.	Suspicious PowerShell execution or brute force attack.	WITHIN 1 HOUR Respond quickly.
MEDIUM 	YELLOW	Moderate risk. Possible threat or suspicious event.	Unusual login from new location or suspicious file.	SAME DAY Investigate today.
LOW 	GREEN	Low risk. Likely benign or informational.	Informational scan, policy change, or expected activity.	MONITOR ONLY No immediate action needed.

RISK IMPACT GUIDE

	HIGH IMPACT ASSETS Domain Controllers, Servers, Databases, Critical Systems = Higher Severity
	NORMAL ASSETS Workstations, User Accounts, Departments = Medium or Low Severity
	LOW IMPACT SYSTEMS Test Systems, Non-Production, Sandbox = Lower Severity

PRIORITY PYRAMID



ANALYST TIPS



- ☒ Don't rely only on severity—always check the **context** and **evidence**.
- ☒ Critical alerts need **immediate** attention—do not delay.
- ☒ **Validate** the alert before taking action.
- ☒ Wrong severity can waste time or cause **missed threats!**
- ☒ When in doubt, **escalate** or ask a senior analyst.



UNDERSTANDING SEVERITY HELPS YOU **PRIORITIZE** BETTER, **RESPOND FASTER** AND **REDUCE RISK!**



17. ALERT TRIAGE FRAMEWORK OVERVIEW



The **ALERT TRIAGE FRAMEWORK** is a step-by-step process that helps you investigate security alerts in a structured and consistent way. It ensures you collect the **right information**, analyze it properly, and **make the right decision**.

THE 4 MAIN STEPS OF ALERT TRIAGE

1 INITIAL ASSESSMENT



- Understand what triggered the alert.
- Identify the affected asset (who/what).
- Check severity and urgency.
- Determine if immediate risk exists.

2 EVIDENCE REVIEW



- Analyze indicators and artifacts.
- Review logs, files, processes, IPs, etc.
- Use tools to verify and gather facts.
- Identify what really happened.

3 CONTEXT GATHERING



- Understand the bigger picture.
- Check user behavior, history, and patterns.
- Look for related alerts or past incidents.
- Determine if activity is normal or unusual.

4 DECISION & ACTION



- Decide: **True Positive** or **False Positive**.
- Take appropriate action.
- Escalate if needed.
- Document and close the alert.

HOW IT WORKS



ALERT RECEIVED



INVESTIGATE
STEP BY STEP



BUILD CONTEXT
& UNDERSTAND



MAKE THE RIGHT
DECISION



DOCUMENT,
CLOSE & IMPROVE

WHY THIS FRAMEWORK MATTERS



- ✓ Keeps your investigation **organized** and **consistent**.
- ✓ Helps you catch **real threats** and reduce **false positives**.
- ✓ Saves **time** and improves **accuracy**.
- ✓ Builds **confidence** and strong analyst decision-making.
- ✓ Makes you a **better SOC analyst** every day!



FOLLOW THE FRAMEWORK. STAY CONSISTENT.
INVESTIGATE **SMART**. MAKE THE **RIGHT CALL**!



18. INITIAL ASSESSMENT



Initial Assessment is your **FIRST LOOK** at the **alert**. You gather the **BASIC DETAILS** to understand what happened, where, how serious, and how urgent it is. This step helps you decide the **NEXT ACTION**.

KEY QUESTIONS TO ANSWER

1 WHAT TRIGGERED THE ALERT?



- What is the alert title?
- Which rule or detection fired?
- What type of activity was detected?

2 WHICH SYSTEM OR ASSET IS AFFECTED?



- What is the hostname?
- Which user or account?
- What type of device or asset is it?

3 WHAT IS THE SEVERITY?



- What severity is assigned?
- Critical, High, Medium, or Low?
- Does it match the risk level?

4 HOW URGENT IS IT?



- Does it need immediate attention?
- Can it wait?
- Is there active risk now?

5 IS THERE IMMEDIATE RISK?



- Is the system compromised?
- Is data at risk?
- Is the attacker still active?

INITIAL ASSESSMENT CHECKLIST



- ✓ Read the alert title carefully – Know what happened.
- ✓ Identify the alert source – Which tool or system generated it?
- ✓ Check the affected host / user – Know the who and what.
- ✓ Look at the severity – Understand the level of risk.
- ✓ Check the timestamp – When did it happen?
- ✓ Assess urgency – Does it need immediate action?
- ✓ Look for obvious threats – Malware, brute force, data exfil?

EXAMPLE – FIRST LOOK AT AN ALERT

Alert Title	Multiple Failed Logins Detected	
Source	Microsoft Defender for Endpoint	
Affected Host	FINANCE-PC03	
User	j.doe	
Severity	High	
Time	May 15, 2025 09:42 AM	
Initial Risk	Possible brute force attack – needs investigation.	



ANALYST TIP

Don't jump to conclusions. Gather the basic details first.

A good first look saves time and prevents mistakes later!



A STRONG INITIAL ASSESSMENT MAKES THE REST OF YOUR INVESTIGATION **FASTER** AND **ACCURATE**!



19. EVIDENCE REVIEW

ANALYZE THE FACTS



The Evidence Review step is where you dig deeper into the data and artifacts. You **verify** what happened, find **key indicators**, and separate normal activity from **suspicious activity**. Good analysis leads to the **right conclusion**.

WHAT TO REVIEW

1

LOGS



- System logs
- Security logs
- Application logs
- Authentication logs
- Firewall / IDS logs

2

ENDPOINT DATA



- Running processes
- File changes
- Registry changes
- Scheduled tasks
- Connections

3

NETWORK ACTIVITY



- Network connections
- DNS queries
- IP address activity
- Web traffic
- Unusual ports

4

ALERT DETAILS



- Alert reason
- Detection rule
- Severity
- Source system
- Timestamps

5

THREAT INTELLIGENCE



- Known threats
- Malicious IPs
- Domains / URLs
- File reputation
- Attack patterns

ANALYSIS CHECKLIST



- ✓ Verify the alert details – What triggered it and why?
- ✓ Correlate events – Do the logs and data match?
- ✓ Identify indicators – What shows suspicious activity?
- ✓ Validate context – Is this normal or abnormal?
- ✓ Look for patterns – Any related activity before/after?
- ✓ Check for impact – What was affected or at risk?
- ✓ Compare with threat intel – Is this a known threat?
- ✓ Document findings – Record what you find clearly.

EXAMPLE – EVIDENCE REVIEW FINDINGS

Category	What We Found	Why It Matters
Logs	Multiple failed logins from 203.0.113.45	Indicates possible brute force attack attempt
Endpoint Data	New suspicious process running (evil.exe)	Could be malware execution
Network Activity	Outbound connection to 185.199.108.153:443	Connection to suspicious external IP
Alert Details	Rule: Possible Credential Brute Force, Severity: High	Confirms high-risk activity
Threat Intelligence	IP is flagged as malicious in threat intel	High confidence this is malicious activity



ANALYST TIP

Don't just collect data – understand it. Look for what's different, unusual, or out of place.

The details tell the real story!



GOOD ANALYSIS TURNS DATA INTO ANSWERS.
ANALYZE DEEPLY. **THINK** CRITICALLY. **STAY** OBJECTIVE!



20.

CONTEXT GATHERING / THE BIGGER PICTURE



Context Gathering helps you understand the bigger **picture**. You collect background information about the **user**, the **system**, the **environment**, and related **events**. This helps you decide if the activity is **normal** or **suspicious**. Context leads to **better decisions**!

KEY AREAS TO GATHER CONTEXT

1

ASSET & USER INFO



- Who is the user?
- What is the system?
- What is its role and function?
- Is it critical?
- Any recent changes?

2

RECENT ACTIVITY



- What was the user doing before?
- Any recent logins or actions?
- Any known issues or maintenance?

3

RELATED ALERTS & INCIDENTS



- Any similar alerts recently?
- Any open incidents related?
- Any patterns or repeated activity?

4

ENVIRONMENT DETAILS



- Location of user or system?
- Network segment or zone?
- Time and timezone?
- Business context?

5

THREAT INTELLIGENCE



- Any known threats targeting this area?
- Malicious IPs, domains, or files?
- Is this activity common for threats?

CONTEXT GATHERING CHECKLIST



- ✓ Identify the user and system involved - **Who and what?**
- ✓ Check recent activity - **What happened before?**
- ✓ Look for related alerts or incidents - **Anything similar?**
- ✓ Understand the environment - **Where, when, and how?**
- ✓ Review threat intelligence - **Any known risks?**
- ✓ Gather all available details - **Build a full picture.**
- ✓ Document your findings - **Keep clear and organized.**

EXAMPLE CONTEXT SUMMARY

 User	j.doe (Finance Department)
 System	FINANCE-PC03 (Workstation)
 Recent Activity	Logged in 10 mins ago, opened Excel, downloaded a file.
 Related Alerts	Similar failed logins from same IP in the last 1 hour.
 Environment	Corporate network, Building A, 10:15 AM (Local Time)
 Threat Intel	IP 185.199.108.153 flagged for brute force attacks.
 Initial Impression	Suspicious activity - needs deeper investigation.



ANALYST TIP

Great context turns data into understanding.

Don't guess—gather facts!

Always ask:

- Who is involved? ✓
- What is happening? ✓
- Where is it happening? ✓
- When did it happen? ✓
- Why could it be happening? ✓



CONTEXT IS THE FOUNDATION OF ACCURATE ANALYSIS.
KNOW THE **STORY**. MAKE THE **RIGHT CALL**!



21.

DECISION & ACTION MAKE THE RIGHT CALL

TRUE POSITIVE

FALSE POSITIVE

ESCALATE



After reviewing the evidence and gathering context, it's time to **make a decision** and **take action**.

Your decision impacts the security of the organization. Be **confident**, be **objective**, and follow the **process**.

POSSIBLE DECISIONS

1 TRUE POSITIVE



The alert is **real** and malicious activity is **confirmed**.

ACTION:

- Contain or block the threat
- Remove or mitigate impact
- Document all actions taken
- Monitor for further activity

2 FALSE POSITIVE



The alert is **not a real** threat. It was triggered by **normal activity**.

ACTION:

- Close the alert
- Document the reason
- Add tuning rules if needed
- Improve detections

3 ESCALATE



Not **enough information** or the alert is **beyond** your authority.

ACTION:

- Escalate to the right team
- Provide all gathered details
- Communicate clearly
- Follow up and track



DECISION & ACTION CHECKLIST

- ✓ Have I collected enough evidence? – **Do I have the facts?**
- ✓ Does the activity match any known threat? – **Is it malicious?**
- ✓ Could this be normal activity? – **Is there a valid reason?**
- ✓ What is the impact if this is real? – **What could happen?**
- ✓ Do I need help or a second opinion? – **Should I escalate?**
- ✓ What action will reduce risk? – **What is the best step?**
- ✓ Have I documented everything? – **Is it recorded clearly?**
- ✓ Have I communicated the decision? – **Is everyone informed?**

DECISION QUICK GUIDE

DECISION	WHEN TO USE	EXAMPLE
TRUE POSITIVE	Malicious activity confirmed with strong evidence.	Malware detected and connection to C2 server confirmed.
FALSE POSITIVE	Alert triggered by normal or expected activity.	Admin running a legitimate script causing an alert.
ESCALATE	Unclear, high impact, or outside your expertise.	Possible data exfiltration but need IR team.



ANALYST TIP

A good decision is based on:

- ✓ **Facts**, not assumptions
- ✓ **Process**, not guesses
- ✓ **Impact**, not just alerts

When in doubt, escalate!



MAKE **SMART** DECISIONS. TAKE THE **RIGHT** ACTIONS. PROTECT YOUR ORGANIZATION. BE A **GREAT** ANALYST!



22.



TRUE POSITIVE vs FALSE POSITIVE

CASE STUDY COMPARISON



Two similar alerts. Two different outcomes.
Let's see how **context** and **investigation** change everything!



TRUE POSITIVE INVESTIGATION



SCENARIO:

Suspicious PowerShell Execution



FINDINGS:



- Encoded PowerShell command
- Connection to external malicious IP (185.199.108.153)
- Malware downloaded from remote server
- User was unaware of the activity
- Threat intel match found (Known malicious IP)



FINAL DECISION:

TRUE POSITIVE



ACTION TAKEN:

- ☒ Isolated the endpoint
- ☒ Terminated malicious process
- ☒ Escalated to Incident Response
- ☒ Reset user credentials
- ☒ Documented all findings and actions

**THREAT CONFIRMED.
ACTION TAKEN.**



FALSE POSITIVE INVESTIGATION



SCENARIO:

PowerShell Alert Triggered



FINDINGS:



- PowerShell script executed by IT admin for maintenance
- Script is approved and stored on internal server
- Connection to known internal server (10.0.1.25)
- Activity performed by IT administrator
- Expected activity during scheduled maintenance window



FINAL DECISION:

FALSE POSITIVE



ACTION TAKEN:

- ☒ Documented findings
- ☒ Closed the alert
- ☒ Added exception to detection rule
- ☒ Notified IT team
- ☒ Improved alert tuning

**NOT A THREAT.
ALERT CLOSED.**



KEY LESSON:
CONTEXT CHANGES EVERYTHING.
INVESTIGATE. ANALYZE. THEN DECIDE.



23. ESCALATION PROCESS



Escalation ensures the **right people** get the right information at the **right time**. When an alert is **complex**, **high-risk**, or beyond your expertise, **escalate** it. Don't guess. **Get help**. **Stay safe!**

ESCALATION WORKFLOW



TIER 1 SOC ANALYST

Initial alert triage and investigation



TIER 2 SOC ANALYST

Deep dive analysis, advanced investigation, validate findings



INCIDENT RESPONSE (IR) TEAM

Contain, eradicate, recover, and conduct threat hunting



SOC MANAGER / LEADERSHIP

Decision making, risk acceptance, reporting, and communication

WHEN TO ESCALATE?



- **High** or **Critical** severity alerts
- Impact is **unclear** or potentially high
- You don't have enough **expertise/tools**
- Malware, data breach, or **active attacks**
- **Multiple** systems or users affected
- Any situation that feels "**not normal**"

WHAT TO PROVIDE?



- Clear summary of the alert
- What happened and when
- Evidence collected
- Indicators (IPs, files, users, etc.)
- Affected assets and users
- Actions already taken
- Your assessment and concerns
- Recommended next steps

WHY ESCALATE?



- Get expert eyes on complex cases
- Reduce risk and prevent damage
- Ensure fast and accurate response
- Follow process and stay within role
- Protect the organization and your team

ESCALATION CHECKLIST



- ☒ Alert summary
- ☒ Evidence collected
- ☒ Severity level
- ☒ Impact assessment
- ☒ Affected systems / users
- ☒ Timeline of events
- ☒ Actions taken so far
- ☒ Recommended actions



ESCALATION TIPS



- ✓ Escalate **early**, not late.
- ✓ Share **clear**, accurate details.
- ✓ Stay **calm** and professional.
- ✓ **Over-communicate** if unsure.
- ✓ Document **everything**.
- ✓ **Follow up** until resolved.



ESCALATION IS **NOT** A WEAKNESS.
IT'S A **SMART** MOVE THAT **SAVES TIME, DATA, AND LIVES!**



24. COMMON BEGINNER MISTAKES



LEARN FROM
MISTAKES.
IMPROVE
EVERY DAY!

Everyone makes mistakes when starting out.
The key is to **recognize** them and **learn**!

1 CLOSING ALERTS TOO QUICKLY

You might miss something important.



HOW TO AVOID:

Take time. Investigate before you decide.

2 IGNORING CONTEXT

Context explains why an alert happened.



HOW TO AVOID:

Always gather context before making a decision.

3 TRUSTING SEVERITY BLINDLY

High severity doesn't always mean real threat.



HOW TO AVOID:

Verify the evidence. Don't trust scores alone.

4 MISSING TIMESTAMPS

Wrong timeline = wrong conclusion.



HOW TO AVOID:

Always check and compare timestamps.

5 NOT CHECKING THREAT INTEL

You might miss known threat information.



HOW TO AVOID:

Check IPs, domains, hashes in threat intel.

6 POOR DOCUMENTATION

Bad notes = confusion for you and your team.



HOW TO AVOID:

Document clearly, completely, and on time.

7 PANIC DURING INVESTIGATIONS

Panic leads to rushed decisions and missed details.

HOW TO AVOID:

Stay calm. Follow the process. One step at a time.



REMEMBER:

Mistakes are normal. Repeating them is **not**.



★ KEY TAKEAWAY ★

Good analysts **learn** from mistakes, stay **curious**, and always try to **improve**.



AWARENESS + **PRACTICE** + **PATIENCE** = **GREAT ANALYST**



25.

REAL ALERT INVESTIGATION
EXAMPLE**SCENARIO: MULTIPLE FAILED LOGINS + SUSPICIOUS IP**

Let's walk through a real-world investigation from start to finish.

INVESTIGATION FLOW (STEP BY STEP)

- 1  **ALERT RECEIVED**
SIEM alert: Multiple failed logins from IP 185.199.108.153
- 2  **CHECK HOSTNAME**
Target host: FINANCE-PC03 (Windows 10 Workstation)
- 3  **REVIEW USER ACTIVITY**
User: j.doe (Finance Department)
Not logged in at the time.
- 4  **ANALYZE IP ADDRESS**
IP: 185.199.108.153
Located in Netherlands
- 5  **CHECK THREAT INTEL**
IP is flagged as malicious in multiple threat intel sources.
- 6  **REVIEW LOGS**
Multiple failed logins using valid username in a short time.
- 7  **DETERMINE SEVERITY**
High risk due to brute force attempts from malicious IP.
- 8  **MAKE DECISION**
All evidence points to a real and active threat.
- 9  **CONTAIN THE THREAT**
Blocked IP, locked account, and isolated the endpoint.
- 10  **DOCUMENT INCIDENT**
Documented all findings, actions taken, and notified the team.

EVIDENCE COLLECTED



Multiple failed logins within 5 minutes



Source IP: 185.199.108.153



Geolocation: Netherlands



Threat Intel: Malicious / High Risk



User not active at the time



Audit logs confirm brute force attempts

ANALYST NOTES

- Multiple failed logins from **malicious IP**.
- User not logged in.**
- IP confirmed as malicious in threat intel.**
- This is not normal behavior.**
- Account likely targeted for brute force attack.



FINAL VERDICT

**TRUE POSITIVE**

This is a real security incident.
Appropriate actions have been taken to contain the threat.

**THREAT
CONFIRMED****KEY TAKEAWAY:** Follow the process. Check the facts. Use context.A methodical approach leads to the **right decision!**

26.

SOC ANALYST
TRIAGE CHECKLIST

Use this checklist **every time** you receive an alert.
Follow the steps. **Don't rush.** **Don't assume.**
Verify. Analyze. Decide. Document.

- | | | | |
|----|--|--|--------------------------|
| 1 | | READ ALERT TITLE
Understand what the alert is about.
Look for key keywords and indicators. | ☐ |
| 2 | | IDENTIFY AFFECTED ASSET
Check the hostname, IP address, user,
and system involved. | ☐ |
| 3 | | CHECK TIMESTAMP
Note the time of the alert and compare
with other related events. | ☐ |
| 4 | | REVIEW SEVERITY LEVEL
Identify severity (Low, Medium, High, Critical)
but don't rely on it blindly. | ☐ |
| 5 | | ANALYZE INDICATORS
Look at IPs, domains, file hashes, URLs,
process names, user agents, etc. | ☐ |
| 6 | | INVESTIGATE CONTEXT
Check user behavior, asset criticality,
location, and recent activity. | ☐ |
| 7 | | VERIFY EVIDENCE
Collect logs, check multiple sources,
and confirm the activity. | ☐ |
| 8 | | DECIDE: TRUE POSITIVE OR FALSE POSITIVE
Based on evidence and context,
make the correct decision. | ☐ |
| 9 | | ESCALATE IF NEEDED
If severity is high or beyond your scope,
escalate to Tier 2 / IR team. | ☐ |
| 10 | | DOCUMENT YOUR FINDINGS
Record steps taken, evidence, decision,
and actions performed. | ☐ |

GOLDEN RULES

**NEVER
ASSUME**

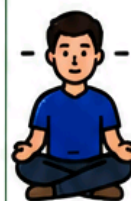
Verify everything
before deciding.

**CONTEXT
MATTERS**

The same alert
can mean
different things
in different
situations.

**EVIDENCE
FIRST**

Decisions should
be based on facts,
not guesses.

**STAY CALM,
STAY FOCUSED**

A calm mind
leads to better
decisions.

**CONSISTENCY
BUILDS SKILL**

Follow the process
every single time.
That's how you
become great!



FOLLOW THE CHECKLIST.
TRUST THE PROCESS. MAKE SMART DECISIONS.
PROTECT YOUR ORGANIZATION!



27. OVERALL SUMMARY



You've learned the **core skills** every SOC analyst needs to **read**, **triage**, and **investigate** security alerts like a pro!

1. ALERTS



- Alerts are signals generated by security tools.
- They help detect suspicious activity.
- Always start every investigation with an open mind.

2. TRIAGE PROCESS



- Prioritize alerts based on severity and context.
- Follow a structured triage workflow.
- Focus on what matters most.

3. TRUE POSITIVE VS FALSE POSITIVE



- True Positive = Real threat.



- False Positive = Benign activity or noise.
- Context and evidence make the difference.

4. INDICATORS OF COMPROMISE (IOCS)



- IP addresses
- Domains
- File hashes
- URLs
- User behavior
- Processes

These indicators help identify threats.

5. SEVERITY LEVELS

CRITICAL

HIGH

MEDIUM

LOW

- Critical = Highest risk
- High = Significant risk
- Medium = Moderate risk
- Low = Minimal risk

Severity helps you prioritize and respond faster.

6. INVESTIGATION WORKFLOW



- Collect evidence
- Analyze context
- Verify indicators
- Determine TP or FP
- Escalate if needed
- Document everything

A consistent workflow leads to accurate results.

WHY IT MATTERS



Faster Response

Detect and stop threats quickly.



Better Decisions

Use data and context, not guesses.



Reduced Risk

Catch real threats and ignore noise.



Stronger Skills

Build confidence and grow your career.

NEXT STEPS



Practice in labs

Hands-on practice builds real skills.



Learn SIEM tools

Master tools like Splunk, Sentinel, QRadar.



Analyze real alerts

The more you analyze, the better you get.



Build your portfolio

Document your work and share your journey.

KEY TAKEAWAYS



Think like an analyst.

Be curious and logical.



Context is everything.

Investigate before you decide.



Evidence first.

Facts over assumptions.



Consistent process.

Follow steps every time.



Great analysts are built through **practice**, **curiosity**, and **consistency**.



28.

FINAL MOTIVATION & BLUETEAMERS CTA!



CONGRATULATIONS, FUTURE SOC ANALYST!

You've completed your guide on how to **read**, **triage**, and **investigate** security alerts. You're building the skills that protect **people**, **data**, and **organizations** every day!



MINDSET OF GREAT ANALYSTS



CURIOUS

Ask questions. Dig deeper. Never stop learning.



PATIENT

Good investigations take time. Rushing leads to mistakes.



LOGICAL

Think critically. Connect the dots. Follow the evidence.



CONSISTENT

Follow the process every time. Small steps create big results.



CALM UNDER PRESSURE

Stay composed. Focus on facts. You make better decisions.

YOUR GROWTH JOURNEY



Keep Learning

Cybersecurity is always evolving. Stay updated. Keep growing.



Keep Practicing

Hands-on practice builds confidence and real-world skills.



Keep Investigating

Analyze more alerts. Learn from every investigation.



Protect & Make an Impact

Your work helps stop attacks and keep the digital world safe.



You're not just reading alerts. You're building a **CAREER** that makes a **REAL DIFFERENCE!**



KEEP GOING. KEEP LEARNING. KEEP DEFENDING.

The best analysts never stop improving!



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